

Speculative Physics and Genetics: Radiation-Induced Mutations from a Quantum Tunneling UFO

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Abstract

This document integrates recent findings from radiation genomics to analyze the biological and genetic consequences of a macroscopic object (a UFO) traveling through the Earth via quantum tunneling or classical penetration at relativistic speeds ($v = 0.90c$). By applying data from Delhomme et al. (2023) on particle radiation and Matsuda & Tanabe (2025) on radiation-induced somatic mutations, we model the catastrophic ionizing radiation environment created by the craft's passage. We quantify the expected DNA damage—specifically non-repeat deletions and clustered mutations—and explore the quantum mechanical phenomenon of proton tunneling in DNA base pairs, demonstrating how the UFO's radiation wake exponentially enhances genetic mutation rates.

1. The Radiation Environment of a Relativistic Craft

A 10,000 kg craft traveling through the Earth's crust, desert sand, or deep ocean at $v = 0.90c$ does not merely displace matter; it annihilates it. The craft is surrounded by a massive plasma sheath, acting effectively as an ultra-high-LET (Linear Energy Transfer) heavy ion.

Using the Bethe-Bloch formula for a macroscopic plasma structure ($Z_{\text{eff}} \approx 10^6$), the energy deposited into the surrounding biological tissue or environment is staggering:

- **Energy Loss:** $\sim 9.3 \times 10^{11}$ MeV/cm
- **Dose at 10 meters:** $\sim 1.5 \times 10^{11}$ Gy

This radiation is composed of a primary Cherenkov photon cone (in transparent media like water) and a massive cascade of secondary particles (gamma rays, neutrons, and secondary electrons) generated by nuclear interactions along the path.

 Figure 1: Dose, Mutation Spectra, and Proton Tunneling

Figure 1: (Left) Radiation dose vs. distance from the UFO path, highlighting lethal and vaporization zones. (Center) Mutation spectra comparing X-rays, proton/alpha particles, and the estimated UFO plasma. (Right) Quantum tunneling probability of protons in DNA hydrogen bonds.

2. Genomic Landscape of Radiation-Induced Mutations

To understand the genetic impact on biological organisms (e.g., humans, deep-sea creatures, or desert flora) that survive the immediate vaporization zone, we turn to recent genomic studies on ionizing radiation.

2.1 The Particle Radiation Signature (Delhomme 2023)

Delhomme et al. [1] investigated the mutational footprints of proton and alpha particle (helium nuclei) radiation on human cells. Unlike X-rays (photons), these heavy charged particles are highly ionizing along their dense tracks.

- **Key Finding:** While the overall rate of Single Nucleotide Variants (SNVs) is not markedly increased, particle radiation produces a distinct increase in **clustered mutations** and structural variants.
- **Mechanism:** The dense ionization tracks of particles cause localized, severe DNA damage that overwhelms the base excision repair (BER) and double-strand break (DSB) repair pathways.

2.2 The Somatic Mutation Landscape (Matsuda & Tanabe 2025)

Matsuda and Tanabe [2] utilized whole-genome sequencing of single cells to map the precise genomic landscape of radiation-induced mutations.

- **Key Finding:** The most distinctive and dose-dependent signature of ionizing radiation is the occurrence of **short, non-repeat deletions** occurring outside of tandem repeats.
- **Secondary Findings:** Structural variants (excluding retroelement insertions) and multisite mutations also show high radiation specificity.

2.3 Application to the UFO Scenario

The radiation wake of the UFO is an extreme version of the particle radiation studied in these papers. The biological tissue exposed to the outer edges of the radiation zone (e.g., at 1,000 to 10,000 meters away) would receive sublethal but highly mutagenic doses.

- **Predicted Mutation Spectrum:** The genome of exposed organisms would be riddled with non-repeat deletions (from the Matsuda signature) and dense clustered mutations (from the Delhomme signature).
- **Cancer Risk:** For a human standing 10 km away, the secondary radiation dose is estimated at ~1.5 Gy (or 30 Sv effective dose for high-LET). Using the ICRP risk coefficient of 5.5% per Sv, the excess lifetime cancer risk approaches 100%, driven by these specific structural deletions and clustered DNA breaks.

 Figure 2: DNA Damage Mechanisms

Figure 2: Infographic detailing the cascade of DNA damage caused by the UFO's radiation wake, incorporating findings from both Delhomme (2023) and Matsuda (2025).

3. Quantum Tunneling of Protons in DNA

While the UFO itself cannot quantum tunnel through the Earth (as proven in previous analyses), quantum tunneling plays a very real and critical role at the genetic level—specifically in the creation of mutations.

3.1 The Löwdin Mechanism

In 1963, Per-Olov Löwdin proposed that protons involved in the hydrogen bonds connecting DNA base pairs (e.g., Adenine-Thymine) could quantum tunnel across the

potential barrier between the bases. If the DNA strands separate for replication while the proton is in the “wrong” position (a tautomeric shift), a permanent mutation occurs.

For a typical hydrogen bond:

- **Barrier Height:** ~ 0.3 eV
- **Barrier Width:** ~ 0.5 Å (5×10^{-11} m)
- **Tunneling Probability:** $T \approx 2.4 \times 10^{-9}$

3.2 Radiation-Enhanced Quantum Tunneling

The intense ionizing radiation from the UFO alters the local quantum mechanical environment of the DNA. Radiation deposits thermal energy and disrupts the electrostatic geometry of the hydrogen bonds, effectively lowering the potential barrier V_0 .

As calculated in our model, if the radiation reduces the effective barrier height by just 20% (from 0.30 eV to 0.24 eV), the quantum tunneling probability increases exponentially:

- **Irradiated Tunneling Probability:** $T \approx 1.2 \times 10^{-7}$
- **Enhancement Factor:** $\sim 50\times$ increase

Therefore, the UFO’s passage not only causes direct kinetic damage to the DNA backbone (DSBs and deletions) but also **catalyzes quantum mechanical mutations** by exponentially increasing the rate of proton tunneling between base pairs.

4. Conclusion

A relativistic UFO passing through the Earth creates a radiation environment characterized by ultra-high-LET plasma interactions and Cherenkov radiation. Based on recent genomic research [1] [2], the surviving biological organisms at the periphery of the event would exhibit a highly specific mutational signature dominated by non-repeat deletions and clustered mutations. Furthermore, the radiation thermalizes the DNA microenvironment, drastically increasing the rate of quantum tunneling-induced tautomeric mutations. The combination of classical radiation damage and enhanced

quantum tunneling would result in catastrophic genetic instability and near-certain carcinogenesis in exposed populations.

References

- [1] Delhomme, T. M., Munteanu, M., Buonanno, M., Grilj, V., Biayna, J., & Supek, F. (2023). Proton and alpha radiation-induced mutational profiles in human cells. *Scientific Reports*, 13(1), 9791. <https://doi.org/10.1038/s41598-023-36845-3> [2] Matsuda, Y., & Tanabe, O. (2025). Genomic landscape of radiation-induced somatic mutations in a normal cell. *Carcinogenesis*, 46(3), bgaf044. <https://doi.org/10.1093/carcin/bgaf044>